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A Specification at a glance

AS Unit 1: The Core Principles of Chemistry

*Unit code 6CH01

- Externally assessed
- Availability: January and June
- First assessment: January 2009

**40%
of the
total AS
marks**

**20% of
the total
GCE
marks**

Content summary:

This unit provides opportunities for students to develop the basic chemical skills of formulae writing, equation writing and calculating chemical quantities. The study of energetics in chemistry is of theoretical and practical importance. In this unit students learn to define, measure and calculate enthalpy changes. They will see how a study of enthalpy changes can help chemists to understand chemical bonding. The study of atomic structure introduces s, p, and d orbitals and shows how a more detailed understanding of electron configurations can account for the arrangement of elements in the periodic table. The unit introduces the three types of strong chemical bonding (ionic, covalent and metallic). Organic chemistry is also introduced with students studying alkanes and alkenes.

Assessment:

Examination of 1 hour 15 minutes in two sections.

Section A: objective test questions.

Section B: mixture of short-answer and extended answer questions.

See *Appendix 5* for a description of this code and all other codes relevant to this qualification.

AS Unit 2: Application of Core Principles of Chemistry***Unit code 6CH02**

- Externally assessed
- Availability: January and June
- First assessment: January 2009

**40%
of the
total AS
marks****20% of
the total
GCE
marks****Content summary:**

This unit develops the treatment of chemical bonding by introducing intermediate types of bonding and by exploring the nature and effects of intermolecular forces. Study of the periodic table is extended to cover the chemistry of groups 2 and 7. Ideas about redox reactions are applied, in particular, to the reactions of halogens and their compounds. The unit develops a largely qualitative understanding of the ways in which chemists can control the rate, direction and extent of chemical change. Organic chemistry in this unit covers alcohols and halogenoalkanes. The treatment is extended to explore the mechanisms of selected examples. Students have to use formulae and balance equations and have an understanding of chemical quantities. Aspects of green chemistry and climate change are also studied.

Assessment:

Examination of 1 hour 15 minutes in three sections.

Section A: objective test questions.

Section B: mixture of short-answer and extended answer questions.

Section C: contemporary context questions.

AS Unit 3: Chemistry Laboratory Skills I***Unit code 6CH03**

- Internally assessed
- Availability: June
- First assessment: June 2009

**20%
of the
total AS
marks****10% of
the total
GCE
marks****Content summary:**

This unit contains practical assessments that cover the content of Units 1 and 2. There is no specific content for this unit. The practical assessments cover the areas of physical, organic and inorganic chemistry. The types of practicals that students must complete are qualitative observations, quantitative measurements and preparations.

Assessment:

Three separate activities testing students' laboratory skills in three different ways. The three activities must cover the areas of physical, organic and inorganic chemistry.

A2**Unit 4: General Principles of Chemistry I – Rates, Equilibria and Further Organic Chemistry*****Unit code 6CH04**

- Externally assessed
- Availability: January and June
- First assessment: January 2010

**40%
of the
total A2
marks****20% of
the total
GCE
marks****Content summary:**

In this unit students make a quantitative study of chemical kinetics and take further their study of organic reaction mechanisms. The topics of entropy and equilibria show how chemists are able to predict quantitatively the direction and extent of chemical change. The organic chemistry in this unit covers carbonyl compounds, plus carboxylic acids and their derivatives. Students are required to apply their knowledge gained in Units 1 and 2 to all aspects of this unit. This includes nomenclature, ideas of isomerism, bond polarity and bond enthalpy, reagents and reaction conditions, reaction types and mechanisms. Students are also expected to use formulae and balance equations and calculate chemical quantities.

Assessment:

Examination of 1 hour 40 minutes in three sections.

Section A: objective test questions.

Section B: mixture of short-answer and extended answer questions.

Section C: data questions, with use of a data booklet.

A2**Unit 5: General Principles of Chemistry II – Transition Metals and Organic Nitrogen Chemistry*****Unit code 6CH05**

- Externally assessed
- Availability: January and June
- First assessment: January 2010

**40%
of the
total A2
marks****20% of
the total
GCE
marks****Content summary:**

In this unit the study of electrode potentials builds on the study of redox in Unit 2, including the concept of oxidation number and the use of redox half equations. Students will study further chemistry related to redox and transition metals. The further organic chemistry section of this unit focuses on arenes and organic nitrogen compounds such as amines, amides, amino acids and proteins. Students are expected to use the knowledge and understanding of organic chemistry that they have gained over the whole GCE in Chemistry when covering the organic synthesis section. This unit draws on all other units within the GCE in Chemistry and students are expected to use their prior knowledge when learning about these areas. Students will again encounter ideas of isomerism, bond polarity and bond enthalpy, reagents and reaction conditions, reaction types and mechanisms. Students are also expected to use formulae and balance equations and calculate chemical quantities.

Assessment:

Examination of 1 hour 40 minutes in three sections.

Section A: objective test questions

Section B: mixture of short-answer and extended answer questions.

Section C: contemporary context questions.

A2 Unit 6: Chemistry Laboratory Skills II***Unit code 6CH06**

- Internally assessed
- Availability: June
- First assessment: June 2010

**20%
of the
total A2
marks****10% of
the total
GCE
marks****Content summary:**

This unit contains practical assessments that cover the content of Units 4 and 5. There is no specific content for this unit. The practical assessments cover the areas of physical, organic and inorganic chemistry. The types of practicals that students must complete are qualitative observations, quantitative measurements and preparations. There is the opportunity for students to undertake a multi-stage experiment, which includes the quantitative measurement and preparation in a longer assessment.

Assessment:

Students' laboratory skills are tested in three different ways, through a choice of activities and pathways. The three activities must cover the areas of physical, organic and inorganic chemistry.

Pathway 1 — as in AS but with practicals covering areas of Units 4 and 5.

Pathway 2 — activity b as in AS, but tasks c and d are together in a multi-stage experiment.

B Specification overview

Summary of assessment requirements

Unit number and unit title	Level	Assessment information	Number of marks allocated in the unit
Unit 1: The Core Principles of Chemistry	AS	The examination will be 1 hour 15 minutes and contain Sections A and B. Section A is an objective test section, and Section B contains a mixture of short-answer and extended answer questions. Section B will include questions on the analysis and evaluation of practical work. Quality of written communication will be assessed in this examination in Section B.	80
Unit 2: Application of Core Principles of Chemistry	AS	The examination will be 1 hour 15 minutes and contain Sections A, B and C. Section A is an objective test section, and Section B contains a mixture of short-answer and extended answer questions. Section C will contain questions on contemporary contexts. This may contain stimulus materials on a scenario that students must read in order to answer the questions. Quality of written communication will be assessed in this examination in either Section B or C. Questions on the analysis and evaluation of practical work will also be included in either Section B or C.	80
Unit 3: Chemistry Laboratory Skills I	AS	Students' laboratory skills will be tested in four different ways, there will be a general practical competence (GPC) check and three separate activities which make up this assessment unit. The four activities must cover the areas of physical, organic and inorganic chemistry. Activities 'b, c and d' include practical activities where the teacher uses Edexcel devised practical activity sheets and mark schemes (if marked by the teacher). Teachers have the option of marking these activities or having them marked by Edexcel. Activity a: General Practical Competence (GPC) This will confirm that students have completed a range of practicals over the whole year and developed their laboratory skills. Students must have carried out at least five practicals in class. The practicals that the students complete must cover the three areas of physical, organic and inorganic chemistry. Verification of completion is required as well as the core practical code (or title), field and dates the practicals were carried out. Activity b: Qualitative observation. Students must complete one qualitative observation, from the selection provided by Edexcel. This activity will include a student sheet, teacher and technician notes and a mark scheme. The activity is marked out of 14. Activity c: Quantitative measurement. Students must complete one quantitative measurement, from the selection provided by Edexcel. This activity will include a	40

Unit number and unit title	Level	Assessment information	Number of marks allocated in the unit
		Activity d: Preparation. Students must complete one preparation (making a chemical), from the selection provided by Edexcel. This activity will include a student sheet, teacher and technician notes and a mark scheme. The activity is marked out of 12. These activities must all be carried out under controlled conditions. Students will be allowed to write up their practical reports in a separate lesson, but their materials must be collected at the end of the session and handed back at the beginning of the next session. Students are not permitted to work on their practical sheets out of the lesson. Students' work must be individual and they may not work with other students in groups. The practical sheets for activities b, c and d are confidential and must not be shown to students prior to their completion of them. They will be available on a secure Edexcel website for teachers to download in preparation for delivery.	
Unit 4: General Principles of Chemistry I — Rates, Equilibria and Further Organic Chemistry	A2	The examination will be 1 hour 40 minutes and contain Sections A, B and C. Section A is an objective test section, and Section B contains a mixture of short-answer and extended answer questions. Section C will contain data questions and will require the use of a data booklet. The longer timing of the examination reflects the style of the Section C questions. Students will be able to show their full ability in Sections B and C as these contain areas where they will be stretched and challenged. They will be provided with data from a laboratory experiment and asked a series of questions on it. Quality of written communication will be assessed in this examination in either Section B or C.	90
Unit 5: General Principles of Chemistry II — Transition Metals and Organic Nitrogen Chemistry	A2	The examination will be 1 hour 40 minutes and contain Sections A, B and C. Section A is an objective test section, and Section B contains a mixture of short-answer and extended answer questions. Questions on the analysis and evaluation of practical work will also be included in Section B. Section C will contain questions on contemporary contexts. This may contain stimulus materials on a scenario that students must read in order to answer the questions. The longer timing of the examination is to reflect the style of the Section C questions. Students will be able to show their full ability in Sections B and C as these contain areas where they will be stretched and challenged. Quality of written communication will be assessed in this examination in either Section B or C.	90
Unit 6: Chemistry Laboratory Skills II	A2	As with AS Unit 3 students' laboratory skills will be tested in four different ways. However there is a choice in how these can be delivered. The laboratory skills assessment must cover the areas of physical, organic and inorganic chemistry.	40

Unit number and unit title	Level	Assessment information	Number of marks allocated in the unit
		All students will have to complete activities a and b as detailed below, as in AS. Activities 'b, c and d' include practical activities where the teacher uses Edexcel devised practical activity sheets and mark schemes (if marked by the teacher). Teachers have the option of marking these activities or having them marked by Edexcel. Activity a: General Practical Competence (GPC) This will confirm that students have completed a range of practicals over the whole year and developed their laboratory skills. Students must have carried out at least five practicals in class. The practicals that the students complete must cover the three areas of physical, organic and inorganic chemistry. Verification of completion is required as well as the core practical code (or title), field and dates the practicals were carried out. Activity b: Qualitative observation. Students must complete one qualitative observation, from the selection provided by Edexcel. This activity will include a student sheet, teacher and technician notes and a mark scheme. The activity is marked out of 14. Students then have a choice of completing either pathway 1 or pathway 2 as detailed below. Pathway 1: Students complete activities c and d, as in AS, as described below. Activity c: Quantitative measurement. Students must complete one quantitative measurement, from the selection provided by Edexcel. This activity will include a student sheet, teacher and technician notes and a mark scheme. The activity is marked out of 14. Activity d: Preparation. Students must complete one preparation (making a chemical), from the selection provided by Edexcel. This activity will include a student sheet, teacher and technician notes and a mark scheme. The activity is marked out of 12. Pathway 2: Students could complete a multi-stage experiment, which comprises a quantitative measurement and a preparation. This covers the same criteria as activities c and d, but gives students experience of a longer and different style practical activity. The activity must be chosen from the selection provided by Edexcel. It will include a student sheet, teacher and technician notes and a mark scheme. This activity is marked out of 26. All of these activities must be carried out under controlled conditions. Students will be allowed to write up their practical reports in a separate lesson, but their materials must be collected at the end of the session and handed back at the beginning of the next session. Students are not permitted to work on their practical sheets out of the	

Information for international centres

A 100 per cent examination option is available **ONLY** to international centres wishing to take GCE Chemistry. This will take the form of alternative written papers for Units 3 and 6.

This option is **not** available for home centres.

International private centres, including all centres entering candidates via the British Council, are not permitted to enter candidates for internal assessment and therefore must take the alternative written papers.

For further details please refer to the Edexcel international website www.edexcel-international.org.

Assessment objectives and weightings

AO1: Knowledge and understanding of science and of How Science Works

Candidates should be able to:

- a recognise, recall and show understanding of scientific knowledge
- b select, organise and communicate relevant information in a variety of forms.

AO2: Application of knowledge and understanding of science and of How Science Works

Candidates should be able to:

- a analyse and evaluate scientific knowledge and processes
- b apply scientific knowledge and processes to unfamiliar situations including those related to issues
- c assess the validity, reliability and credibility of scientific information.

AO3: How Science Works

Candidates should be able to:

- a demonstrate and describe ethical, safe and skilful practical techniques and processes, selecting appropriate qualitative and quantitative methods
- b make, record and communicate reliable and valid observations and measurements with appropriate precision and accuracy
- c analyse, interpret, explain and evaluate the methodology, results and impact of their own and others' experimental and investigative activities in a variety of ways.

		% in AS	% in A2	% in GCE
AO1	Knowledge and understanding of science and of <i>How Science Works</i>	38.4%	25%	31.7%
AO2	Application of knowledge and understanding of science and of <i>How Science Works</i>	38.4%	43%	40.7%
AO3	<i>How Science Works</i>	23.2%	32%	27.6%
TOTAL		100%	100%	100%

Relationship of assessment objectives to units

Unit number	Assessment objective			
	AO1	AO2	AO3	Total for AO1, AO2 and AO3
Unit 1	10%	8%	2%	20%
Unit 2	8%	10%	2%	20%
Unit 3	1.2%	1.2%	7.6%	10%
Unit 4	5.3%	9.4%	5.3%	20%
Unit 5	6%	10.9%	3.1%	20%
Unit 6	1.2%	1.2%	7.6%	10%
Total for Advanced Level GCE	31.7%	40.7%	27.6%	100%

Qualification summary

Subject criteria

The General Certificate of Education is part of the Level 3 provision. This specification is based on the Advanced Subsidiary GCE and Advanced Level GCE subject criteria for chemistry which are prescribed by the regulatory authorities and are mandatory for all awarding bodies.

The GCE in Chemistry has been designed to build on the knowledge and skills set out in the GCSE Science criteria, and provide further opportunities for students to consider the applications and implications of chemistry and the development of scientific ideas.

The GCE in Chemistry criteria includes *How Science Works*, which is explained below in more detail. It also includes experimental and investigative skills, mathematical requirements and key skills.

Aims

The aims of the Edexcel Advanced Subsidiary GCE and Advanced Level GCE in Chemistry are to develop:

- students' interest in, and enthusiasm, for chemistry, including developing an interest in further study and careers in chemistry
- an appreciation of how society makes decisions about scientific issues and how the sciences contribute to the success of the economy and society
- a deeper understanding of the skills, knowledge and understanding of *How Science Works*
- essential knowledge and understanding of different areas of the subject and how they relate to each other.

How Science Works

This Edexcel Advanced Subsidiary GCE and Advanced Level GCE specification requires students to develop the skills, knowledge and understanding of *How Science Works*, which are described as follows:

- use theories, models and ideas to develop and modify scientific explanations
- use knowledge and understanding to pose scientific questions, define scientific problems and present scientific arguments and ideas
- use appropriate methodology, including ICT, to answer scientific questions and solve scientific problems
- carry out experimental and investigative activities, including appropriate risk management, in a range of contexts
- analyse and interpret data to provide evidence, recognising correlations and causal relationships
- evaluate methodology, evidence and data and resolve conflicting evidence
- appreciate the tentative nature of scientific knowledge
- communicate information and ideas in appropriate ways using appropriate terminology
- consider the applications and implications of science and appreciate their associated benefits and risks
- consider ethical issues in the treatment of humans, other organisms and the environment

-
- appreciate the role of the scientific community in validating new knowledge and ensuring integrity
 - appreciate the ways in which society uses science to inform decision making.

These statements have been embedded within the specification and will be assessed both in the examinations and as part of the internal assessment. A more detailed mapping of where these statements are found within the units can be found in *Appendix 6*. This also contains a further expansion on the specification points which address these *How Science Works* statements.

Experimental and investigative skills

This Edexcel Advanced Subsidiary GCE and Advanced Level GCE specification requires students to develop a range of experimental and investigative skills. These are highlighted in the previous *How Science Works* section.

Core practicals that all students must carry out have been listed within the unit specification, at both AS and A2 level. Many of these will be standard GCE practicals. This ensures that all students cover a variety of different practicals and have developed their experimental and investigative skills. The examinations can then contain questions of a more practical nature, such as analysis and evaluation of practical work, discussion of errors, safety procedures and risk assessments.

The GCE in Chemistry criteria assessment objective 3 (AO3): *How Science Works*, states that students should be able to:

- demonstrate and describe ethical, safe and skilful practical techniques and processes, selecting appropriate qualitative and quantitative methods
- make, record and communicate reliable and valid observations and measurements with appropriate precision and accuracy
- analyse, interpret, explain and evaluate the methodology, results and impact of their own and others' experimental and investigative activities in a variety of ways.

These points have been embedded within the specification and the internal and external assessments, both at AS and A2 level.

With all laboratory practicals it is essential that centres carry out a detailed risk assessment before allowing students to carry out the practical. For further information on risk assessments and chemical hazards please refer to the CLEAPSS website (www.cleapss.org.uk).